SIMULATING LOWER LIMB PROSTHETIC USE FOR DATA-DRIVEN CARE

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Abstract— Prosthetic Component Selection is a labor-intensive process due to the absence of a public repository for real-world performance data. Numerous prosthetic components are on the market, and no public unified database of component ratings; prosthetists face challenges in identifying optimal solutions. Although intellectual property (IP) is available within larger companies, prosthetists often use patients' rating feedback for post-fitting; however, a "cold start" problem exists for new users and newly introduced components, which thus limits the efficiency. Machine learning offers an opportunity to address these challenges by uncovering latent trends and similarities to provide algorithmic recommendations across components, categories, and outcome measures. This paper proposes a PySpark-powered, case-based recommender that leverages outcome and satisfaction scores to recommend optimal Feet/Knee/Hip components. The system reduces time and research by clinicians and increases fitting accuracy.

Keywords—Prosthetic components, PySpark, Outcome-based recommendation

I. INTRODUCTION

Prosthetic component selection involves choosing the right parts to create a functional and comfortable artificial limb. This process requires careful consideration of the amputees' needs, lifestyle, and physical capabilities, as well as the expertise of prosthetists. With the absence of public access databases of prosthetic components and real-world performance data, prosthetists rely on patients' feedback post-fitting and intellectual property (IP) of components within larger companies, which is inefficient and inconsistent with new users and new components introduced to the market. We believe an algorithm rooted in ML may assist in faster optimization of prosthetic component selection. This study aims to provide optimal recommendations for Feet/Knee/Hip components using a PySpark-powered, case-based recommender that leverages outcome and satisfaction scores.

II. DATA ACQUISITION AND CLEANING

Because no publicly available dataset exists, we simulated 1M patient cases to approximate real-world distributions. Python and the 'random' and 'fake' modules were utilized to engineer features and populate the dataset (see Table 1). Patient identification (patient_id) was generated to prevent duplication. Age values were randomly distributed between 18 and 90 years. Prosthetic components were sampled at random from three major manufacturers and assigned using domain knowledge (Figure 1). Amputation levels were allocated according to real world clinical incidence rates of lower-limb amputations (transtibial, transfemoral, and hip disarticulation). Both amputation side and gender were randomly assigned with equal probability ($p = 0.5$), resulting in a balanced 50/50 distribution for each. Follow-up visits were generated randomly using a Poisson probability with an average rate of 4 since domain knowledge would have patients visiting on average 1 to 2 times per year. Delivery date was randomly set to a date that includes up to 3 years prior. The outcome score represents a normalized measure of the patient's functional performance with the device (Figures 2 and 3), while the satisfaction rating reflects overall satisfaction on a Likert scale. Feature engineering includes normalizing visit frequency over a clinically meaningful timeframe (e.g., visits per six months) and combining the outcome score and satisfaction rating into a single weighted performance metric.

Table 1. Columns and descriptions of dataframe contents

Column Name	Description
patient_id	unique patient identifier
age	patient age (18 years <= age <= 90 years)
gender	patient gender (male/female)
amputation_level	level of amputation of lower limb (transtibial, transfemoral, or hip disarticulation)
amputation_side	Side of the body of the amputation (right/left)
foot_type	Foot component (28 possible components)
knee_type	Knee component (27 possible components)
hip_type	Hip component (3 possible components)
fitting_date	Date the prosthesis with the component(s) was fit/delivered (< 3 years prior)
num_visits	Follow-up visits since delivery/fitting_date
outcome_score	Normalized outcome score of patients (0 – 100)
satisfaction_rating	Likert scale of patient satisfaction (1 – 5)

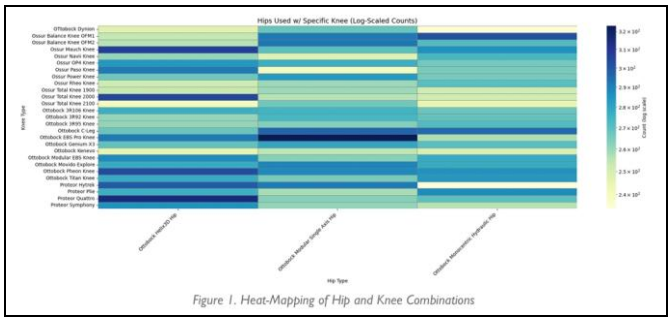


Figure 1. Heat Mapping of Hip and Knee Combinations

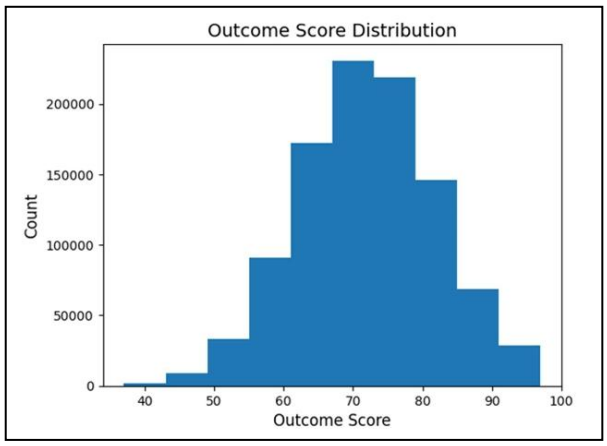


Figure 2. Distribution of Outcome Scores in Dataset

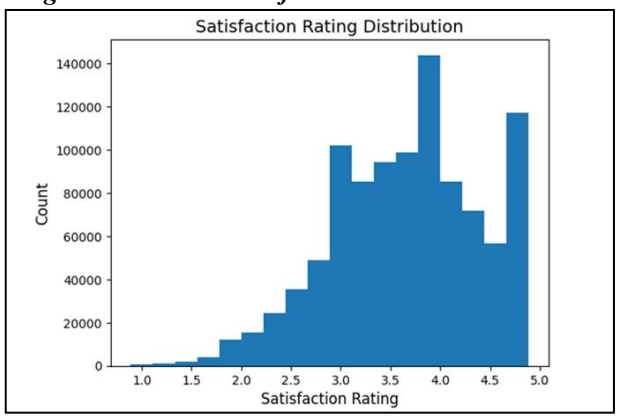


Figure 3. Distribution of Satisfaction Rating in Dataset

III. METHODS

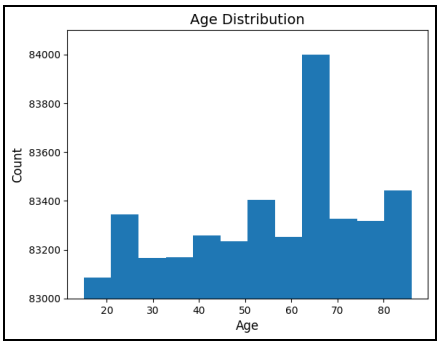
Due to the size of the dataset, PySpark is applied as a distributed cluster framework for data processing. Pipelines are constructed to handle both pre-processing tasks and modeling. A knowledge-based recommender system (RS) is developed to assemble a recommendation set for the user.

In the first stage, case-based filtering is performed using specific information provided by the user. A memory-based model (k-NN) then computes similarities to identify N neighbors. These neighbors are aggregated, and re-ranking is performed based on outcome and satisfaction scores to prioritize the highest-scoring options. Finally, the system returns the top N (e.g., 3 or 5) recommendations relevant to the clinician's case scenario.

To assist in building trust in the RS, additional feedback can be provided to the prosthetist, such as explanations for why a recommendation was made. The system will be evaluated using RMSE (for score/rating prediction accuracy) and Precision@k (e.g., 3 or 5) to measure recommendation accuracy on hold-out datasets.

IV. EXPLORATORY DATA ANALYSIS

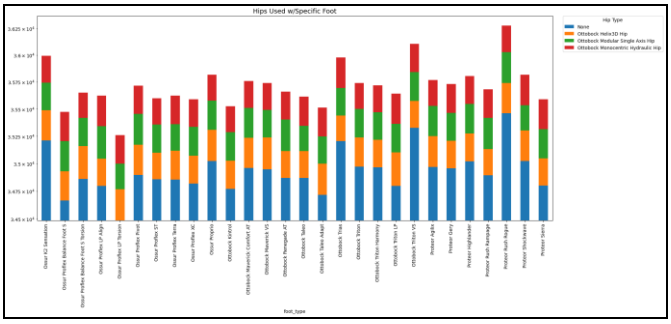
The team conducted an exploratory data analysis (EDA) on the dataset to address the following research questions: (1) What are the demographics of the amputees? Figures 4 and 5 show that individuals aged 60–70 represent the largest group in the dataset, followed by those over 80 and those aged 50–55. A smaller peak is also observed in the 20–25 age range. Figure 5 indicates that males stand a slightly higher proportion of the dataset compared to females. (2) Are there correlations between hips used and specific knees/feet? Figures 6 and 7 present frequency reports showing how often particular hips appear in combination with specific knee or foot components. (3) Are there correlations between knees used and specific feet? Figure 8 illustrates the frequency with which different knees appear together with specific foot components. (4) What are the correlations between follow-up visits and age, satisfaction scores, and outcome scores? A correlation analysis (Figure 9) was conducted across these factors to help assign weights for subsequent modeling.



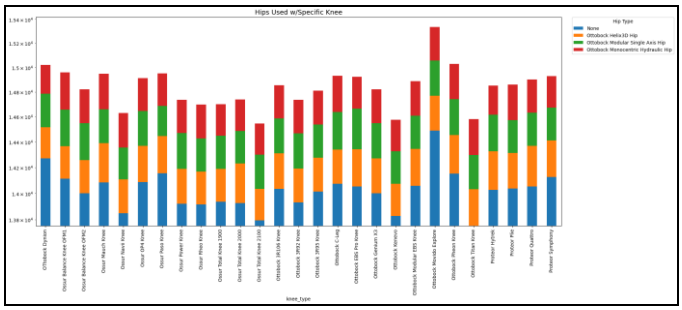
Figures 4. Distribution of Age in Dataset

```
+------+------+
|gender|count |
+------+------+
|m     |500280|
|f     |499720|
+------+------+
```

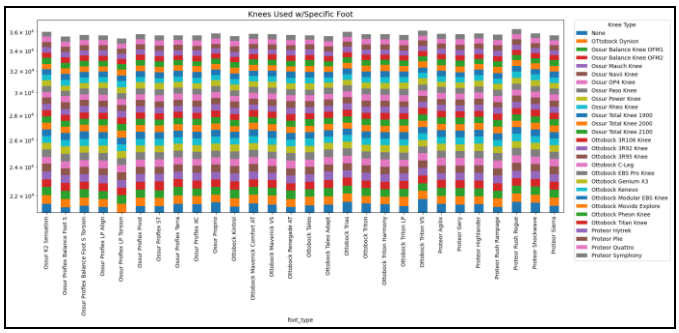
Figures 5. Distribution of Gender in Dataset



Figures 6. Hip Used with Specific Foot



Figures 7. Hip Used with Specific Knee



Figures 8. Knee Used with Specific Foot

```
# -------Check correlations of numeric cols-------

from pyspark.ml.feature import VectorAssembler
from pyspark.ml.stat import Correlation

num_cols = ['age', 'outcome_score', 'satisfaction_rating', 'num_visits']

assembler = VectorAssembler(inputCols=num_cols, outputCol="feat_vec")
vec_df = assembler.transform(sdf).select("feat_vec")

# compute correlation matrix
correlation_matrix = Correlation.corr(vec_df, "feat_vec", method="pearson").head()[0]
print(correlation_matrix.toArray())
```

```
[[ 1.00000000e+00  2.85745851e-04 -1.94626984e-03  1.91289723e-03]
 [ 2.85745851e-04  1.00000000e+00 -1.06351133e-04  1.54878555e-03]
 [-1.94626984e-03 -1.06351133e-04  1.00000000e+00 -9.95320175e-05]
 [ 1.91289723e-03  1.54878555e-03 -9.95320175e-05  1.00000000e+00]]
```

Figures 9. Correlation Analysis of follow-up visits/age/satisfaction scores/outcome scores

V. MACHINE LEARNING MODELING

The team applied PyShark for machine learning process. First step we create a user interface to gather input of patients, including age, gender, amputation side, and the amputation level (Figure 10). We identified columns of interest into component, category, numerical, and demographic, then imported PyShark ML features for setting up vectors, transforming, following with pipeline and transformation into SDF (Figure 11). In "weighted_rating" column, we applied 20% the frequency of follow-up as 20% of the weight and 40% each for the outcome score, and the satisfaction rating in the weighted sum. After we transformed category features into usable vector, we performed BucketedRandomProjectionLSH to get neighbors then get nearest neighbors pool and order by similarity before we sort the result to conclude "top neighbors" then returned to user (Figure 12).

```
What age is the patient?:
18
What is the patient's gender?: M/F
M
What side is the amputation?: L/R
A
Please enter either  "R"  or  "L"
What side is the amputation?: L/R
L
What is the amputation level?: Transtibial/Transfemoral/Hip Disarticulation
Transfemoral
```

Figure 10. User Interface for Input of Patients

Figure 11. and Figure 12 attached as appendix below

VI. RESULT AND FUTURE ENGINEERING

The findings demonstrate that our system can generate "top neighbors" as recommended prosthetic components based on patient-provided inputs through the interface (Figure 13). This approach holds considerable potential to advance the field, particularly with future developments that integrate additional variables influencing prosthetic selection, such as healthcare coverage, occupational demands, and socioeconomic factors. Moving forward, we intend to engage with institutions such as Penn Medicine and Independence Blue Cross to obtain access to data that is currently unavailable and to benefit from their professional expertise. Through such collaborations, we can take an important step forward toward creating a more efficient and patient-centered process for prosthetic component selection.

```
The top recommended feet based on your selections and ordered by
highest rating are:

        -Ottobock Triton Harmony
        -Ottobock Maverick VS
        -Ottobock Maverick Comfort AT

The top recommended knees based on your selections and ordered by
highest rating are:

        -Ottobock Kenevo
        -Ossur Paso Knee
        -Proteor Quattro
```

Figure 13. Return Recommendation to Users

References

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[5] Roy, D., Dutta, M. A systematic review and research perspective on recommender systems. *J Big Data* **9**, 59 (2022). https://doi.org/10.1186/s40537-022-00592-5

APPENDIX

patient_id	features_norm	weighted_rating	foot_type	knee_type	hip_type	gender	amputation_side	amputation_level	age
62961957	(11,[0,4,6,10],[0...	0.9136854523613159	Ottobock Renegade AT	None	None	m	r	transtibial	69
50063462	(11,[1,4,6,10],[0...	0.7790140059413846	Ottobock Taleo Adapt	None	None	f	r	transtibial	39
49975743	(11,[0,4,6,10],[0...	0.7637815062914867	Ossur Proflex LP ...	None	None	m	r	transtibial	59
71491108	(11,[0,4,7,10],[0...	0.8451286081111522	Ossur Proflex Bal...	Ottobock Genium X3	None	m	r	transfemoral	72
56539946	(11,[0,3,6,10],[0...	0.726	Ossur Proprio	None	None	m	l	transtibial	26

only showing top 5 rows

Figure 11. Apply transformation to sdf

patient_id	features_norm	weighted_rating	foot_type
30660285	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.8696935978047251	Proteor Rush Rampage
30270024	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.717316346137909	Proteor Rush Rampage
18821615	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.7561610621767199	Ottobock Triton VS
78518743	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.8237274171568063	Ossur Proflex Terra
53728481	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.7430851089381902	Ossur Proflex Balance Foot S
29814958	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.7363541233428436	Ossur Proflex Terra
51246168	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.8702751130917726	Ossur Proflex ST
20372731	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.8663051805277484	Ossur Proflex Balance Foot S
22337075	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.8774104292571883	Proteor Gery
25191916	(11,[0,3,7,10],[0.410880752562212,0.410880752562212,0.410880752562212,-0.70251763075508])	0.8454856586309918	Ottobock Maverick Comfort AT

only showing top 10 rows

knee_type	hip_type	gender	amputation_side	amputation_level	age
Ossur Paso Knee	None	m	l	transfemoral	18
Ottobock Movido Explore	None	m	l	transfemoral	18
Ossur OP4 Knee	None	m	l	transfemoral	18
Ottobock C-Leg	None	m	l	transfemoral	18
Ottobock Pheon Knee	None	m	l	transfemoral	18
Ossur OP4 Knee	None	m	l	transfemoral	18
Ottobock C-Leg	None	m	l	transfemoral	18
Ossur Paso Knee	None	m	l	transfemoral	18
Ossur Power Knee	None	m	l	transfemoral	18
Ottobock 3R95 Knee	None	m	l	transfemoral	18

age	hashes	distCol
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0
18	[[1.0], [-3.0], [0.0], [0.0], [-1.0], [0.0], [-1.0], [-1.0], [0.0], [-1.0], [-1.0], [1.0], [-2.0], [2.0], [-1.0], [-1.0]]	0.0

Figure 12. Nearest Neighbors Pool and Order by Similarity